Climbing the Ladder: Automating Data Pipelines and Reproducing Baselines in Low-Resource Settings

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Abstract

The massively multilingual translation systems that now include Sicilian learned it from noisy, non-standardised web text. To obtain better translation quality, we redevelop the small neural machine translation (NMT) system that Wdowiak [35] customized for Sicilian. We rebuild on maintained software the extraction of the parallel text from the PDFs of the bilingual journal *Arba Sicula*, whose sentences are aligned by a language-agnostic sentence encoder with thresholds tuned against an existing hand alignment and then filtered for the typical errors of automatic alignment. Furthermore, the tokenization and the desinence-biased subword segmentation of the original system are ported to a Perl-free implementation, and all the results are measured on a frozen test set drawn from the journal. We compare small Transformers trained from scratch with the pretrained NLLB model adapted by low-rank fine-tuning, in both directions and with Italian as a closely related bridge language, and we adapt to the pretrained model the reverse-training recipe of the original system, i.e., a stage on forward-scored back-translations of the web Sicilian followed by a stage on the curated literary text. The pretrained model is by far the better of the two, and the staged training improves the translation into Sicilian more than the translation from it. On the test set of the original system, and in the tokenized space in which it was scored, the staged model with a larger adapter and more back-translations is above the reverse-training recipe in the directions that involve English and below it between Sicilian and Italian. Yet, a multilingual adapter degrades the directions on which it was not trained, which limits its use as a general multilingual model.

1 Introduction

Languages without an army and navy often lack a standard orthography. Without specific spelling rules and conventions, written texts in these languages often exhibit great variability (sometimes even within a local dialect), which inhibits the ability of conventional metrics to assess model performance [1]. But even languages like Sicilian – where published dictionaries have presented a consistent orthography for several hundred years [14, 15, 18, 31] – can exhibit extreme orthographic diversity which (when incorporated into a training dataset) prevents a

model from learning how to generate consistent text and often causes the model to generate foreign words instead of native words.

During the 13th century, the Sicilian School of Poets at the court of Emperor Frederick II created the first literary standard in Italy. Their poetry inspired Dante to write in his native Florentine, which later became today's Italian language. Following in their footsteps, poets like Antonio Veneziano (1543-1593), Giovanni Meli (1740-1815), Domenico Tempio (1750-1821), Ignazio Buttitta (1899-1997) and Salvatore Camilleri (1921-2021) continued the Sicilian literary tradition, writing in a fairly consistent manner. But writing is different from speaking, so Giuseppe Pitrè (1841-1916) developed a popular tradition which recorded the common people in their own words and in the sound of their own words.

One can hear echoes of the popular tradition in the daily speech of millions of people in Sicily, Calabria and Puglia, as well as in diaspora communities abroad. And one can find echoes of the literary tradition in books, journals and pages of the internet. Nonetheless, the publicly available Sicilian language text often exhibits extreme orthographic diversity which once prevented practitioners from developing datasets of Sicilian-language parallel text and once caused Sicilian to be absent from every major translation service.

Sicilian is now covered by massively multilingual models such as NLLB-200 [16]; however, its training text was mined from non-standard web sources, and the model has in effect learned a non-standard orthography (cf. section 4). Hence, the problem for Sicilian has moved from the absence of a sufficiently large dataset (on which to train a model) to the quality of available training data.

High-quality material exists, especially in the bilingual literary journal *Arba Sicula* (published since 1979), which Wdowiak [35] used to create the first Sicilian neural machine translation (NMT) system. His work demonstrated that it's possible to develop a fairly good translation model in low-resource settings, but it was a labor-intensive task. He aligned sentences by hand using a recipe tailored to the low-resource, morphologically rich setting.

This work rebuilds that effort with maintained, mostly CPU-only tooling and seeks to obtain better quality with modern pretrained models.

That system [34, 35] followed the low-resource recipe of Sennrich and Zhang [25], i.e., a small and heavily regularized Transformer [32] with a small subword vocabulary [27], and biased the segmentation toward the desinences of Sicilian grammar; its later reverse-training strategy [36] builds a pretrained model of Sicilian in stages from synthetic and web text before the fine-tuning on the literary one.

Traditionally, parallel text was automatically aligned with dictionary-based tools like hunalign. More recently, practitioners have begun using multilingual sentence encoders like LaBSE [6] and LASER3 [7], which were used to mine WikiMatrix [23] and the NLLB's Sicilian data. Vecalign [29] combines such embeddings with a dynamic program, and collections such as OPUS [30] gather the resulting corpora.

Pretrained multilingual models can be adapted to a specific pair at a small cost by low-rank adapters [9], and allow one to exploit closely related high-resource languages as a bridge [2, 5, 10, 37]; also, back-translation of monolingual text [26] is the standard way of adding in-domain synthetic data. Section 2 explains how one can use such methods to overcome the challenges of low-resource translation, as originally described by Koehn and Knowles [11].

Here, we extract the parallel text of the journal automatically, with a LaBSE-based aligner whose thresholds are tuned against the second author's hand-aligned text. The aligner's output is then filtered to remove any remaining errors. And during training, the automatically aligned text and the hand-aligned sheets are deduplicated and used together, because the latter cover only a fraction of the pairs of each issue.

Furthermore, we port to Python the tokenization and the desinence-biased segmentation of the original system, and add a lemma source factor [24] to the from-scratch baseline. Finally, we fine-tune NLLB-200 with LoRA in both directions and with Italian as a bridge, and we replace

the custom pretraining of the reverse-training recipe with the pretraining of NLLB, keeping its second stage (forward-scored back-translations of the web Sicilian) and its last one (the curated literary text). Since the published results of the original system were measured on a different test set and in the space of its tokenizer, we score our models on that test set and in that space as well (cf. Table 5).

The structure of the paper is as follows. Section 2 recalls the models on which all our systems are based. Section 3 describes the extraction, the alignment and the cleaning of the parallel text and the assembly of the datasets, while section 4 discusses the orthography of the Sicilian data and the reason why a multilingual encoder can align it at all. Tokenization and subword segmentation are the subject of section 5, the models and the staged training of section 6, and the evaluation protocol of section 7; the results are reported in section 8, and section 9 concludes.

2 Background

We now recall the model on which all our systems are based. NMT treats translation as a conditional language model: an encoder reads the source sentence $x = (x_1, \dots, x_n)$, and a decoder factorizes the probability of the target sentence y from left to right as

$$p_\theta(y \mid x) = \prod_t p_\theta(y_t \mid y_{<t}, x) \ ,$$

where the parameters θ are trained by maximizing the log-likelihood of a parallel corpus, and the translation is obtained by beam search. Its standard parameterization is the Transformer [32], which replaces recurrence with the scaled dot-product attention $\mathrm{Attention}(Q, K, V) = \mathrm{softmax}(QK^\top/\sqrt{d_k})V$ (where Q, K and V are the query, key and value matrices, and d_k is the dimension of the keys), computed in several parallel heads.

With little data a deep neural network overfits, so we implement the best practices of Sennrich and Zhang [25] who recommend a smaller model with high dropout [28] and a small subword vocabulary. Byte-Pair Encoding (BPE) [27] gives an open vocabulary and shares statistics across the forms of a word, which is especially useful for a morphologically rich language such as Sicilian. Further improvements usually come from deeper models, when the data allow them [13], and from linguistic input features [24], which we use in section 6.

3 Data Pipeline

3.1 Extraction from PDFs

We read the text layer of the *Arba Sicula* PDFs with PyMuPDF, which avoids the pdflatex round trip of the original pipeline. However, the quality of this layer depends on the issue. In fact, the PDFs of the issues up to AS18, and of AS21, are scans of the paper copies whose OCR text carries no accent over the vowels; from AS19 (1998) onward they were produced from the original computer files. This date also marks (most likely) an improvement of the Sicilian itself, since 1998 is the year of Privitera's *Beginner's Sicilian* [22], the first Sicilian grammar published after the one of Pitrè [19], which was followed by the more comprehensive grammar of Bonner [3]; hence, the scanned issues are kept out of the training data (they are used only for the complete extraction of the archive mentioned below). Some of the born-digital issues (e.g., AS22 and AS23) contain spaces of zero width inside the words, which we detect from the glyph positions and remove before the text is read.

Each page is assigned a language according to its share of Sicilian and English stop-words, and pages with too few tokens are left out. Since the journal usually sets Sicilian and English on facing pages, each Sicilian page is compared with the English page on either side, and the

pairs are chosen greedily by decreasing LaBSE similarity (cf. section 3.2), each page being used at most once; this corrects the pairings that a fixed left/right rule gets wrong wherever a page is mislabelled or the order of the two languages is swapped. Pages are then segmented into sentences with `pysbd` (with the Italian rules for Sicilian), after the running headers and page numbers have been removed; fragments with fewer than two words or mostly in capitals are dropped, and the words hyphenated at the end of a line are rejoined.

3.2 Sentence alignment

We embed each sentence with LaBSE [6], a language-agnostic dual encoder, and build the cosine-similarity matrix

$$S_{ij} = \frac{\langle u_i, v_j \rangle}{\|u_i\| \, \|v_j\|} \, ,$$

where u_i and v_j are the embeddings of the i-th Sicilian and of the j-th English sentence. For a pair of pages, the similarity is the larger of the mean row maximum and the mean column maximum of S, and the pair is kept if it is at least τ_{page} (with at least two sentences on each page). Consecutive spreads are then chained into blocks, so that a sentence cut by a page break is mended when it has no closing punctuation and the next page starts in lowercase. Within a block, the sentences are aligned by a monotonic dynamic program over S with moves 1–1, 1–2 and 2–1, whose gain is the mean cosine of the sentences involved, and moves 1–0 and 0–1, which cost a penalty of 0.5; the program is banded to the facing page and its neighbors (and falls back to the unbanded one if the band admits no path). An aligned span is kept if its mean cosine is at least τ_{sent}. This is in the spirit of Vecalign [29], but requires neither a bilingual dictionary nor hunalign.

Both thresholds were chosen on a grid (0.50, 0.55, 0.58 and 0.62 for τ_{page}, 0.35, 0.40, 0.45 and 0.50 for τ_{sent}) against the hand alignment of the second author for the issues AS41 and AS42, which gave $(\tau_{\text{page}}, \tau_{\text{sent}}) = (0.50, 0.40)$ (its F1 of 0.58 is 0.01 below that of $\tau_{\text{sent}} = 0.35$, which keeps more pairs at a lower precision). On that gold the aligner reproduces 715 pairs exactly (recall 0.54, precision 0.63); these figures are somewhat pessimistic, since a pair that is correct but segmented differently from the hand alignment counts as an error, and since the hand alignment favored the pairs that were easier to align. Each pair records the printed Sicilian and English page and its LaBSE similarity, which allows one to check it against the printed issue.

3.3 Cleaning

A review of the aligned text showed that it's easy to identify the aligner's errors, which makes LaBSE's alignments much easier to perfect than hunalign's. Because LaBSE's errors are easy to identify, we filter them as follows. First, the residue of the page layout is removed from the sentences (the InDesign slug of AS43 and the "Return to the TOC" links). Then, a pair is dropped if (i) the longer side has more than 1.6 times the words of the shorter one (when it has at least five words), (ii) it contains a table-of-contents leader, (iii) it contains two or more characters typical of OCR noise, or (iv) more than 25% of its lowercase English words are not in an English dictionary. Note that the bound in (i) was first 2.0; the pairs between 1.6 and 2.0 have a mean LaBSE similarity of 0.59, against 0.73 for those below 1.6, which in our view justifies the tighter bound. After cleaning and deduplication, the whole digital archive of the journal gives 15,465 pairs.

3.4 Datasets

Two training sets are used (Table 1). Our first set, on which the from-scratch models and the first NLLB experiments were trained, joins our first extraction of 44 issues (14,245 pairs, before

Table 1: Datasets (number of sentence pairs). "Web" is the first training set of section 3.4, "curated" the second one; the test set is shared.

set	direction	pairs	sources
web	scn–en	27,386	Arba Sicula (11,538) + Good-Sicilian-in-NLLB (15,848)
	it–scn	9,389	WikiMatrix
curated	scn–en	30,980	hand-aligned sheets, Arba Sicula, Napizia
	scn–it	10,427	hand-aligned sheets, Napizia
	scn (mono)	14,360	Cipolla's second textbook
test	scn–en	1,000	Arba Sicula
	it–scn	1,000	WikiMatrix

the cleaning of section 3.3) with the "Good-Sicilian-in-NLLB" set curated by the second author, i.e., sentences mined with LASER3 [7, 16] that were rescored and filtered for quality and for Corsican contamination. Its it–scn part is the WikiMatrix subset [23], with a small curated "Good-Sicilian-from-WikiMatrix" set.

Our second set, used by the staged training of section 6.3, replaces the web text with curated sources. These are (i) the hand-aligned sheets of the second author, i.e., Dieli's translation of Pitrè's fables and 23 issues of the journal (scn–en), Cipolla's translations of Meli (scn–it), and the trilingual exercises of the grammars of Bonner and Cipolla, (ii) our alignment of all the born-digital issues (AS19 to AS47, except AS21), since the sheets cover only 10 to 20% of the pairs of an issue and the two sources are therefore complementary, and (iii) the parallel pages of the Young Sicilian Manifesto, of the Napizia website and of the Napizia Magazine, from which one text under copyright was excluded. Furthermore, the sheets provide 14,360 monolingual Sicilian sentences (Cipolla's 2021 textbook) and a validation set of 121 trilingual lines. Its Sicilian side is normalized as in section 5, and the exact repeats are removed.

A validation and a test set of 1,000 pairs each were drawn once at random from our first extraction of *Arba Sicula*, which provides the literary standard that FLORES does not, among the pairs with 3 to 80 words per side and a length ratio between 1/3 and 3, and then frozen; hence, all the scn–en results of section 8 are computed on the same test set, except those on the test set of the second author (Tables 5 and 6, and the lower half of Table 3). The validation set of 1,000 pairs is used with the web set and to choose the epoch of the runs of Table 3, whereas the 121 lines of the sheets are the test set of the second author (cf. section 7), and are never used for model selection, since the staged runs of section 6.3 train for a fixed number of epochs. Each training set is then guarded against it: a pair is removed if its Sicilian side is a test sentence (in raw or normalized form) or, in the second set, contains a test sentence of six or more words; the validation pairs are removed in the same way from the web set and from the runs of Table 3, and the 121 lines of the second author from the second set. This guard removed 794 pairs from the second set (778 because of the test set and 16 because of the 121 lines of the second author), some of them from the hand-aligned sheets, which cover the same issues as the test set; without it, the results of the staged training would have been inflated.

4 Standardization

Although Sicilian does not have a single official orthography, a common literary standard has existed for several hundred years and provides a rough consistency, which we can refine because the majority of the Sicilian sentences in the NLLB dataset were written in that standard.

Nonetheless, approximately one-third of the Sicilian sentences that the NLLB team collected were written in local dialect. Some of the sentences were written in Corsican. And very few of the Sicilian sentences in the NLLB dataset were paired with a valid translation. We suspect

that these data quality issues were caused by seed data which is inconsistent with the literary standard.

4.1 Why Sicilian lives in the Romance neighbourhood

A pretrained multilingual encoder needs no explicit Sicilian support to *read* Sicilian. The representations it learns are not a shared interlingua that maps every language into one abstract, universal meaning space; instead they cluster languages by linguistic similarity [12] and organise around surface-form cues [33]. For Sicilian and Italian, which are close on both counts, the two accounts agree: Sicilian lands in the Italian–Romance region either way.

The surface form of literary Sicilian is close to Italian, so reading it works well. Retrieval on the multi-parallel FLORES-200 devtest [16] confirms it: LaBSE retrieves the correct English translation of a Sicilian sentence in 99% of the cases (precision@1), one point below Italian (100%), which it supports officially. Furthermore, if one discounts English (the pivot of LaBSE, to which all languages are close), the nearest translation of a Sicilian sentence is Italian in 49% of the remaining cases, more than twice as often as French (21%), the next Romance language.

The seed-data failure, however, is in the *opposite* direction – retrieving the Sicilian translation of an English sentence. Romance languages share a lot of vocabulary and grammar, so when a model retrieves a Sicilian, Italian or Corsican translation of an English sentence, the translation that it selects will reflect the surface-form cues in its training data. Sometimes the NLLB seed data's surface form reflects the literary language in Sicilian dictionaries, but other times it reflects local pronunciations, which (we suppose) is what caused their model to frequently retrieve Corsican-language sentences when assembling the NLLB's Sicilian corpus.

Since multilingual models encode script and surface form more than deeper linguistic structure [33], the two directions behave differently for the same reason: an Italianate surface form retrieves the right English sentence, whereas seed data that look like Corsican lead the model to generate the wrong Sicilian. Hence, the proximity to Italian cuts both ways. It allows our LaBSE-based aligner (cf. section 3.2) to work on a language that LaBSE does not officially support; however, it gives no protection against contamination by neighboring languages when a corpus is assembled by transfer only.

How easily a text is aligned depends, in turn, on how standard it is. We computed, text by text, the LaBSE cosine of the hand-aligned gold pairs of AS41 and AS42: standard literary prose (Cipolla's *Don Chisciotti*) has a mean cosine of 0.76, with 91% of its pairs above $\tau_{\text{sent}} = 0.4$, while dialect poetry has a mean of 0.64, with 80% of its pairs above it. Hence, the non-standard material is both the larger source of noise in the collected data and the harder to recover, and curating the data toward the literary standard should improve quality on both counts. For poetry, we also tried merging short consecutive verses into longer units before scoring. We simulated it on the poetry pairs of the gold: merging up to three verses raises the mean cosine by about 0.11 and the share above the threshold by 15 points (from 0.65 and 84%, pooled over all the poetry pairs, to 0.76 and 99%), although part of this gain is mechanical, since the merged units are fewer, longer and more distinctive. This corrective is not yet part of the production aligner.

For a translation model, we need sentences written in the Sicilian literary language and we need translations of those sentences. So we back-translated all of the Sicilian sentences into English and Italian, then we scored the resulting pair on the task of translation into Sicilian. Because a sentence written in standard Sicilian will score better than a sentence written in local dialect, forward-scoring the back-translations allows us to identify the sentences written in standard Sicilian. This method yielded 549,486 back-translations to simulate English-to-Sicilian translation and 689,840 to simulate Italian-to-Sicilian translation.

5 Tokenization and Subword Segmentation

Because the majority of the Sicilian sentences in the NLLB dataset were written in a common literary standard, we can refine them for consistency. We first normalize the text toward the standard established by Cipolla and augmented by Wdowiak for MT clarity, i.e., strict use of L on definite articles (*lu/la/li*), Italian-style H on *aviri* verbs (*haiu, hai, havi, avemu, aviti, hannu*), CI for çi (*çiuri*→*ciuri*) and R retained before a clitic (*farlu*). Our normalizer applies the documented spelling rules (e.g., *cchiù*→*chiù*, *io/iu/eu*→*jo*, *non*→*nun*) at a "std" level, and further uncontracts and ASCII-folds the text at a "full" level. Since the tokenizer of NLLB was trained on un-normalized web Sicilian, it is not obvious that normalization helps the pretrained model; hence, it was measured on scn→en, with the same English references for the three variants. BLEU is 30.67 on raw text, 30.79 with "std" and 30.61 with "full" (chrF 56.31, 56.49 and 56.36, respectively). Thus, the light "std" normalization gives the best scores, although by only 0.1 BLEU, which is within the noise of repeated runs, while the "full" one is no better than raw text, which is consistent with a pretrained tokenizer that expects un-normalized input. We therefore adopt "std" for all the later experiments.

For the from-scratch models, the Napizia tokenizer was reimplemented in Python (its output coincides with that of the original Perl code, except that it also normalizes dashes and ellipses): it ASCII-folds the accents and uncontracts the Sicilian forms (e.g., ô→*a lu*, dû→*di lu*), which reduces sparsity. Then, the joint scn–en BPE [27] is biased toward the textbook desinences catalogued in the resources that Wdowiak assembled to standardize Sicilian (Dieli's *Sicilian Vocabulary*, Cipolla's *Mparamu lu sicilianu* [4], Bonner's grammar [3] and the *Chiù dâ Palora* dictionary), which are appended to the Sicilian side when the merges are learned. This keeps the morphological endings as units; for instance, Wdowiak's sentence *Carinisi are dogs!* is segmented as `car@@ in@@ isi are dogs !`. In his experiments this reduced the vocabulary from 14,000 to 2,000 units and gave the largest improvement in BLEU. We call this combination of tokenizer and biased segmentation "lever B" in the following.

6 Models

6.1 From-scratch baseline

The from-scratch baseline is a small Transformer trained on CPU with Sockeye-3 [8]: 3 layers, model size 256, 4 attention heads, feed-forward size 1024, a vocabulary shared by the two languages and a joint BPE of 4000 merges, i.e., about 6.7M parameters. Dropout is 0.4 on the embeddings and 0.2 elsewhere, with label smoothing 0.1, and the model is trained with Adam (learning rate $2 \cdot 10^{-4}$, batches of 1024 words) for at most 30 epochs, with early stopping after 8 checkpoints without improvement. Pairs with a side longer than 100 subwords are left out of training. On top of it we add, in this order, lever B and a linguistic input feature ("lever D"). The latter is a lemma source factor [24] summed into the source embedding as

$$e_j = E_{\text{word}}(w_j) + E_{\text{lemma}}(\ell_j) \,,$$

where w_j is the j-th source word and ℓ_j its lemma; the lemma vocabulary is small (about 900 entries), and therefore generalizes across the inflected forms of a word. The lemmas are those of the lemma table of the original system, which maps 16,377 inflected forms (mostly of verbs) to 1,252 lemmas; a word missing from the table receives an unknown lemma.

6.2 Pretrained model

Our pretrained model is NLLB-200 [16], a massively multilingual encoder–decoder that supports Sicilian (`scn_Latn`) and selects the target language through the first token forced on the decoder.

We adapt its 1.3B variant with LoRA [9]: the pretrained weights W_0 of the query, key, value and output projections of the attention are frozen, and a low-rank update is learned, which gives $W = W_0 + \frac{\alpha}{r}BA$ with r=32, α=64 and dropout 0.05 on the adapter (about 18.9M trainable parameters, i.e., 1.4% of the model). As usual, the base model is kept in fp16 and the adapters in fp32; training uses a learning rate of $2 \cdot 10^{-4}$ and sentences of at most 128 tokens, and decoding uses beam search with 5 beams (4 for the synthetic sentences below). One adapter is trained on both directions at once (scn→en and en→scn), and one on four (scn↔en and it↔scn). The former is trained for two epochs, the latter for one. For the back-translation experiment of section 8, we translate into Sicilian with the bidirectional adapter the 7,592 English sentences left unaligned by the first extraction, keep the 7,498 synthetic pairs whose length ratio is between 0.5 and 2 and whose Sicilian side is new, and train on them and on the web set a further adapter for scn→en, for one epoch, on top of the bidirectional one (merged into the base model).

6.3 Staged training

To obtain a pre-trained model, we can either develop a custom one from scratch or we can download one from a model hub. In this subsection, we explore both. To build a custom pretrained model, we trained one in stages. First, we trained a bidirectional Italian–English model on ParaCrawl, then we further trained the same model on Sicilian forward-translations of that text. In the next stage, we trained on back-translations from the NLLB's Sicilian text. And in the final stage, we fine-tuned on *Arba Sicula*. We can similarly further train the NLLB-200 in stages, but because the model already provides a multilingual pretraining, we only need to perform the last two stages, which require data prepared for this task (distinct from the back-translations mentioned in section 4). Starting from the monolingual Sicilian of NLLB (1,057,469 sentences), we back-translated it into English and Italian with the Sockeye models, and scored each synthetic pair by the negative log-probability that its real side is a translation of the synthetic one (the lower, the better); since his models were trained on the literary standard, a standard Sicilian sentence usually obtains a lower score than a dialectal one. We did the same with the English side of the NLLB data (translated into Sicilian), with 941,391 Italian sentences of WikiMatrix (translated into Sicilian) and with the monolingual Sicilian of his sheets. Our second stage takes, for each of these files, the 100,000 pairs with the best score (all of them for the two files of his sheets, which have fewer) among those that pass a few filters: no unknown token, at least five words on each side, a source that is neither equal to nor a near-copy of the target (similarity below 0.9), no overlap with the test set, and a Sicilian side that does not end its words in -o more often than in -u, and has at most one word in -o (an inexpensive test against Italian passing as Sicilian, which tends to obtain the best scores). Together, the six lists cover the directions scn→en, en→scn, it→scn and scn→it, and 100,000 natural it–en pairs of WikiMatrix are added in both directions; the adapter is trained on them for one epoch. Finally, the third stage continues the same adapter on the curated set of section 3.4 in the four directions scn↔en and scn↔it, for two epochs.

Furthermore, we also experimented with not discarding the pairs with the worst scores, which contain the dialectal, Italian and Corsican text that the second stage filters out. In these experiments we spent them in a first, coarse stage, which spreads the output probabilities across the dialects and leaves broad peaks roughly in the right places. We then allowed the later stages to sharpen them. This curriculum prepends to the two stages above one epoch on the 200,000 worst-scoring pairs of each file (all of them for the two smaller files) (with at least three words on each side and no filter on the endings), and is otherwise identical to them, so that the two runs can be compared.

7 Evaluation

We score the systems with sacreBLEU [21], i.e., with BLEU [17] (tokenization `13a`) and chrF [20], on the frozen literary test set of section 3.4. We do not use the FLORES test set of NLLB for scn–en, since its Sicilian follows the orthography introduced after 2017 and comes from a different, non-literary domain, and is therefore not representative of the *Arba Sicula* material; FLORES-200 devtest is used only for it↔en, which our test sets do not cover. Sockeye systems are scored in tokenized space, as the upstream system was, whereas NLLB systems are scored on raw text, with the Sicilian side normalized as in training (except for the out-of-the-box rows marked as raw); the two groups of rows of Table 2 are thus not strictly comparable (the floor obtained by copying the source is 5.18 BLEU in the tokenized space and 5.44 on raw text). Training was done on an 8-core CPU for the Sockeye models and on a single Colab GPU for the NLLB adapters. We report no running times, since only the quality of the translation is of concern here.

For the comparison with the published results of the second author (cf. Table 5), the NLLB systems are also scored on his test set in the six directions between Sicilian, English and Italian. This set consists of the 121 trilingual lines that he selected by hand from the issues AS38 and AS39 of the journal (labeled as validation in his sheets). As he did, we compute BLEU in the space of the Napizia tokenizer of the original system, i.e., on the tokens that it produces from the hypothesis and from the reference (accents folded and contractions undone, with the Italian side going through the English tokenizer). This space is more lenient into Sicilian, since an accent or an apostrophe placed differently no longer costs a match: on our test set it raises the en→scn score of the staged model of section 6.3 from 21.72 to 25.69 BLEU, and its scn→en score only from 32.37 to 33.03. Since some of his lines also occur in our training data (8 in that of the staged runs and 14 in the web set), we report the scores both on all 121 lines and on the lines absent from the training data of each model. The re-decoding of a model reproduces its BLEU only up to a few hundredths (e.g., 32.37 against 32.40 on scn→en), since beam search in fp16 on the GPU is not exactly deterministic.

8 Results

Table 2 reports the results on the frozen test set of 1,000 literary pairs. In the table, "floor" is the score of the untranslated source, each row beginning with "+" adds one component to the row above it, "raw" means that NLLB receives the Sicilian text without normalization, and a dash marks a direction on which the model was not trained or evaluated. Note that the Sockeye baseline exceeds the floor by only 3.3 BLEU (8.51 against 5.18, both in the tokenized space), that lever B raises it to 10.94, and that the lemma factor of lever D lowers it to 10.18. NLLB-1.3B, on the other hand, is much better even out of the box (29.00 on scn→en), and the bidirectional LoRA adapter trained on the web set raises it to 31.27, while en→scn improves from 9.90 to 19.19. Adding the 7,498 back-translated pairs gives 31.60 on scn→en; this small improvement is consistent with the size of the in-domain monolingual pool, which is two orders of magnitude smaller than the back-translations used by the staged training.

Staged training improves both directions, with 32.40 BLEU on scn→en and 21.72 on en→scn; the gain over the bidirectional adapter is 1.13 BLEU on scn→en and 2.53 on en→scn, which may be due to the back-translations having been selected for the quality of their Sicilian, i.e., of the target side of en→scn. We remark that this comparison mixes two changes, since the staged run replaces the web set with the curated one as well as adding the back-translations; we have not trained the curated set alone. Prepending the coarse stage of section 6.3 raises scn→en to 33.01 BLEU (58.62 chrF), 0.61 more than the staged run, while en→scn is essentially unchanged (21.78 against 21.72). Repeated runs of the bidirectional adapter differ by up to about 0.2 BLEU on scn→en and 0.5 on en→scn, and therefore the first difference is probably

Table 2: Results on our test set (BLEU / chrF). The upper rows are trained on the web set, the staged rows on the back-translations and on the curated set (cf. section 6.3).

model	scn→en	en→scn
floor (copy source)	5.18 / 25.95	–
Sockeye baseline	8.51 / 32.73	–
+ lever B (tok. + desinences)	10.94 / 34.66	–
+ lever D (lemma source factors)	10.18 / 34.15	–
NLLB-200 1.3B, out of the box (raw)	29.00 / 55.23	9.90 / 41.09
LoRA, bidirectional	31.27 / 56.92	19.19 / 50.02
+ back-translation (7,498 pairs)	31.60 / 57.21	–
staged (back-translations, then curated)	32.40 / 58.19	21.72 / 52.79
curriculum (coarse stage first)	33.01 / 58.62	21.78 / 52.94

real and the second is within the noise (however, the repetition of the staged run in Table 3 differs from it by 0.42 BLEU on scn→en, which makes the first difference less certain). Of course, the curriculum also trains for one more epoch on about 826,000 pairs, and hence we cannot tell whether its benefit comes from the dialectal text itself or from the longer training.

Ablation of the staged training. In order to find which settings limit the staged model, we have run it again with the settings of section 6.3 (base) and with three changes made together. These are (i) 300,000 pairs per file in the second stage instead of 100,000, (ii) an adapter of rank 64 instead of 32 (with α=128) that also covers the feed-forward layers, and (iii) sequences of up to 256 tokens instead of 128 (at 128 tokens, about 2% of the curated pairs are truncated), also in decoding, with three epochs in the third stage instead of two. All runs hold out of training our test and validation sets and all the lines of the test set of the second author, in the three languages. The epoch of the third stage is chosen on the 1,000 validation pairs, and only that epoch is scored on the two test sets (Table 3). The base run reproduces the staged run of Table 2 within 0.42 BLEU (32.82 and 21.52 BLEU against 32.40 and 21.72), and the three changes raise it to 35.40 BLEU on scn→en and 25.09 on en→scn, i.e., by 2.6 and 3.6 BLEU. On the test set of the second author, the gain is 3.9 BLEU into Sicilian, 3.8 from it and 3.5 from Italian into Sicilian, while scn→it gains only 0.6. The two directions between English and Italian lose 1.8 and 2.2 BLEU instead; they are not trained in the third stage, and the longer stage possibly makes them forget more. The larger second stage alone gives 33.34 and 21.88 BLEU on our test, i.e., 0.5 and 0.4 more than the base run, which is about the noise of repeated runs; on the test set of the second author it gains 1.4 BLEU from Sicilian into English and 2.7 from Italian into Sicilian, and changes the other directions by about 1 BLEU at most. Adding the larger adapter gives 34.60 and 24.18 BLEU on our test, i.e., 1.3 and 2.3 more than the larger second stage alone, and the longer sequences with the third epoch add a further 0.8 and 0.9; hence, most of the gain comes from the adapter. On the test set of the second author, however, the variant without the longer third stage is better in five directions out of six (e.g., 62.02 against 60.95 BLEU from Sicilian into English, and 61.89 against 59.09 from Italian into English), and it is worse only from English into Sicilian (47.64 against 48.36). Since the variants are compared on the validation set, which favors the longer third stage (33.17 and 23.28 BLEU against 32.54 and 22.07), we keep the latter in Table 5. Finally, since it→en and en→it are not trained in the third stage, we have also added to it 10,427 natural WikiMatrix it–en pairs per direction that the second stage did not use (last column), reusing the same second stage. On our test the scores do not change (35.49 and 25.16 BLEU), and on the test of the second author scn→it gains 0.9 BLEU (62.94), but it→en and en→it do not recover (57.30 and 47.53

Table 3: Ablation of the staged training (cf. section 6.3): the base settings, the larger second stage alone, the larger second stage with the rank-64 adapter, three changes made together (300,000 back-translated pairs per file, a rank-64 adapter on the attention and feed-forward layers, 256-token sequences and three epochs in the third stage), the latter with 10,427 natural it–en pairs per direction in the third stage, and the final model, i.e., the three changes with the WikiMatrix it–scn pairs in the third stage and up to four epochs. The epoch is chosen on the validation set; our test is scored on raw text (BLEU / chrF), his on all 121 lines in his tokenized space (BLEU), since all runs exclude them from training.

	base	+ 300k pairs	+ rank 64	+ three changes	+ it–en	final
epoch chosen	2 of 2	2 of 2	2 of 2	3 of 3	3 of 3	4 of 4
our test						
scn→en	32.82 / 58.41	33.34 / 58.75	34.60 / 59.89	35.40 / 60.39	35.49 / 60.41	36.01 / 60.70
en→scn	21.52 / 52.68	21.88 / 52.97	24.18 / 54.70	25.09 / 55.24	25.16 / 55.09	26.34 / 55.75
test of the second author						
en→scn	44.42	44.49	47.64	48.36	48.46	49.25
scn→en	57.14	58.54	62.02	60.95	59.81	59.34
it→scn	59.31	62.04	64.42	62.78	61.86	61.19
scn→it	61.46	61.02	62.67	62.05	62.94	60.30
it→en	60.90	59.97	61.89	59.09	57.30	58.42
en→it	50.16	49.14	48.76	48.00	47.53	48.77

against 59.09 and 48.00), possibly because the encyclopedic text of WikiMatrix is far from the literary one of his test. Since its validation score is slightly lower (33.19 and 22.89 BLEU), this variant is not kept. Also, the validation score was still rising at the third epoch (23.28 BLEU into Sicilian against 22.06 at the second), which suggests that a longer third stage may improve the model further. The final model therefore keeps the three changes, adds to the third stage the 9,389 WikiMatrix it–scn training pairs of the web set in both directions (whose validation and test pairs stay out) and allows up to four epochs, reusing the same second stage (column "final"). Its validation score still rises at the fourth epoch (33.58 and 23.83 BLEU), the highest of all the variants, and on our test it reaches 36.01 BLEU on scn→en and 26.34 on en→scn, i.e., 0.6 and 1.3 more than the three changes. On the test of the second author, however, it gains only from English into Sicilian and into Italian (49.25 against 48.36, and 48.77 against 48.00), and it loses 0.7 to 1.8 BLEU in the other directions, among which the two between Sicilian and Italian that the WikiMatrix pairs were meant to help (61.19 and 60.30 against 62.78 and 62.05). Since it is the best variant on the validation set, it is the model of Table 5.

Italian directions. Italian is closely related to Sicilian and has much more data, and NLLB supports it↔scn natively. On the web set, we hold out a frozen it–scn test set of 1,000 pairs from the WikiMatrix data, evaluate NLLB-1.3B out of the box, and then fine-tune the four directions (scn↔en and it↔scn) jointly into one multilingual adapter, which uses Italian as a bridge in the sense of Johnson et al. [10] and Fan et al. [5]. Table 4 reports the results as BLEU / chrF. As one would expect for a close, high-resource neighbor on in-domain WikiMatrix text, the Italian pairs are by far the "easiest": scn→it reaches 43.47 BLEU, and it→scn improves from 17.57 to 26.97. However, the multilingual adapter loses somewhat on scn↔en (30.75 and 17.49 BLEU, against 31.27 and 19.19 for the bidirectional adapter of Table 2); hence, on the web set the Italian bridge brings no improvement to the English directions.

Forgetting of the untrained directions. Since the adapter is trained only on the directions that involve Sicilian, one may ask what happens to the others. On FLORES-200, the trilingual adapter lowers it→en from 34.42 to 27.14 BLEU and en→it from 29.42 to 28.77 (Table 4), i.e.,

Table 4: Trilingual adapter on the web set, out of the box and after fine-tuning (BLEU / chrF). The scn–en rows use our test set, the it–scn rows the WikiMatrix one, and the it–en rows the FLORES-200 devtest, on which the adapter was not trained.

direction	out of the box	multilingual FT
scn→en	29.00 / 55.23	30.75 / 56.57
en→scn	9.90 / 41.09	17.49 / 48.78
it→scn	17.57 / 44.76	26.97 / 51.69
scn→it	42.23 / 59.05	43.47 / 59.79
it→en	34.42 / 62.65	27.14 / 57.92
en→it	29.42 / 58.63	28.77 / 58.07

it forgets a direction on which the base model was good and on which it was never trained. This is the transfer from high- to low-resource languages described by Johnson et al. [10], and it limits the use of the adapter as a general multilingual model; the translation between Sicilian and the other languages of NLLB (e.g., Spanish, French or Portuguese) is probably affected as well. On the FLORES-200 devtest, the trilingual adapter gives 7.8 to 11.4 BLEU into Sicilian from these three languages and 19.1 to 30.1 BLEU from Sicilian into them; however, we have not scored the base model on these directions, and hence we cannot tell how much of it is lost.

Relation to the state of the art. Table 5 compares our NLLB systems with the published results of the second author on his test set and in the tokenized space in which he scored them (cf. section 7). Its first three columns are his, i.e., the baseline and the larger many-to-many (m2m) model of Wdowiak [35] (the latter trained with back-translations and, for the Italian directions, with the trilingual textbook exercises), and the third stage of his reverse-training system [36]. The other columns are ours, i.e., the bidirectional adapter trained on the web set (the adapter of Table 2, whose re-decoding scores 31.30 and 19.19 BLEU on our test set against 31.27 and 19.19) and the two staged runs of section 6.3, each scored on all 121 lines and on those absent from its training data. On the latter, the staged model is on a par with the reverse-training system from English and from Italian into Sicilian (45.07 against 45.1 BLEU, and 61.49 against 61.4), and it exceeds it by 7.8 BLEU from Sicilian into English and by 2.0 and 14.3 BLEU between English and Italian; it remains below it only from Sicilian into Italian (59.84 against 62.9). The baseline of 2022 (25.1 on en→scn and 29.1 on scn→en) is exceeded by about 20 and 27 BLEU, and already the web adapter exceeds it by 6.9 and 21.6 BLEU. The curriculum is somewhat better than the staged run on scn→en (58.19 against 56.43) and somewhat worse on en→scn (43.92 against 45.07), which on 113 lines is probably within the noise. The final model of the ablation (cf. Table 3) raises the score into Sicilian to 49.25 BLEU and the one from it to 59.34, i.e., 4.2 and 10.7 BLEU above the reverse-training system; it remains below it from Italian into Sicilian (61.19 against 61.4) and from Sicilian into Italian (60.30 against 62.9), and it loses somewhat between English and Italian w.r.t. the staged run. The gap of Table 2 between our en→scn score and his baseline (21.72 against 25.1) may therefore be due to the test set and to the metric rather than to the model: on raw text, the staged model obtains 39.43 BLEU into Sicilian on the clean lines of his test set and 21.72 on ours, and the tokenized space adds a further 3 to 6 BLEU into Sicilian (cf. section 7).

It is fair to remark, however, that the comparison is not entirely even. The baseline of the second author chose its checkpoints on this same test set (training stopped when the score on it ceased to improve), and so did his later systems, whereas our staged runs train for a fixed number of epochs and were scored on these lines only after training. On the other hand, his training data do not contain the test lines (although they contain other text of the same articles), whereas ours do. Also, his test file exists in several revisions, which are only intermediate stages

Table 5: Published results of the second author against ours on his test set, in his tokenized space (BLEU). His columns are taken from Wdowiak [35] and Wdowiak [36]; in ours, the first number is the score on all 121 lines and the second the score on the lines absent from the training data of that model (107 for the web adapter, 113 for the staged runs); the last column is the final model of Table 3, which was trained without any of the 121 lines and is therefore scored on all of them. The web adapter was trained on scn↔en only, and a dash marks a direction that was not reported.

	Wdowiak			ours (NLLB-200 1.3B)			
direction	baseline	m2m	reverse	web	staged	curriculum	final
en→scn	25.1	35.0	45.1	32.19 / 31.95	44.40 / 45.07	43.97 / 43.92	49.25
scn→en	29.1	36.8	48.6	51.16 / 50.65	57.00 / 56.43	58.80 / 58.19	59.34
it→scn	–	36.5	61.4	39.69 / 39.50	61.04 / 61.49	60.69 / 60.86	61.19
scn→it	–	30.9	62.9	52.98 / 52.57	60.40 / 59.84	60.47 / 59.96	60.30
it→en	–	–	48.2	56.59 / 55.63	62.72 / 62.45	61.38 / 61.11	58.42
en→it	–	–	46.7	47.31 / 45.89	49.27 / 48.69	49.14 / 48.62	48.77

of its tokenization. Finally, the first stage of his recipe, i.e., the adaptation of an Italian–English model to Sicilian, is the part that we have not reproduced; on his test set its role seems to be taken by the multilingual pretraining of NLLB, with the possible exception of scn→it, the only direction in which his system remains ahead.

Reproduction of the original baseline. In order to separate the effect of the model from that of the data, we have also retrained the 2022 baseline of the second author with his configuration, as recorded in his repository. This uses Napizia tokenization, separate BPE vocabularies of 3,000 merges per language (the Sicilian one learned with the inflections of the Dieli and Chiu da Palora vocabularies appended), and a Transformer with 3 layers, model size 256, 4 heads and feed-forward size 1,024, whose target embeddings are tied to the output layer. Dropout is 0.5 on the embeddings and 0.25 elsewhere, with label smoothing 0.1, and the model is trained with Adam (learning rate $1.5 \cdot 10^{-4}$, batches of 20 sentences) for at most 20 epochs, with Sockeye 3 instead of his Sockeye 1. Pairs with a side longer than 200 subwords are left out, as in his configuration (his later many-to-many model used 100). We train it on three sets, i.e., (A) his hand-aligned sheets, (B) our automatic alignment of the issues from AS19 to AS47 with the textbook exercises, and (C) both, deduplicated, each guarded against his test set and against the validation lines, and we score it on his test set in the tokenized space, as in Table 5; Table 6 reports the results.

On his data, the reproduction remains well below the published scores (15.97 against 25.1 BLEU on en→scn, and 16.65 against 29.1 on scn→en). Two differences of protocol may account for at least part of the gap: he started each model from a previously trained one rather than from scratch, and he stopped the training on the test set, whereas we stop on 500 lines of our validation set. Also, none of our six runs converged within the 20 epochs (the validation score was still improving when the cap was reached), which suggests that a longer training, such as the one implied by his warm start, would raise them. Our alignment (B) gives 2.8 to 3.5 BLEU less than his sheets, which is consistent with its smaller size and with the absence of Dieli's translations of Pitrè; however, the two together (C) give 22.45 and 20.69 BLEU, i.e., 6.5 and 4.0 more than A, and therefore the automatic alignment contributes pairs that the hand alignment does not contain. Of course, all these scores remain far below those of the NLLB adapters on the same lines (Table 5).

Table 6: The 2022 baseline of the second author retrained from scratch with his configuration, on three training sets, and scored on his test set in his tokenized space (BLEU). The first row is the published score of Wdowiak [35], obtained with a warm start and with the training stopped on the test set.

training set	pairs	en→scn	scn→en
published (his sheets)	–	25.1	29.1
A (his sheets)	20,220	15.97	16.65
B (our alignment)	18,503	13.20	13.14
C (both)	31,730	22.45	20.69

9 Conclusion and Future Work

We have rebuilt, on maintained software and without Perl, the path from the PDFs of *Arba Sicula* to a Sicilian–English NMT system, i.e., the extraction, alignment and cleaning of the parallel text, the tokenization and desinence-biased segmentation of the original system with an added lemma source factor, and the adaptation of NLLB-200 with LoRA, trained in stages on the back-translations of the second author and on the curated literary text. On our test set the best model, i.e., the staged training with three times the back-translations, a larger adapter, longer sequences and the WikiMatrix it–scn pairs, reaches 36.01 BLEU on scn→en and 26.34 on en→scn, while the coarse stage proposed by the second author helps the translation from Sicilian, but not the one into it. On the test set of the second author and in his tokenized space, the same model reaches 49.25 BLEU into Sicilian and 59.34 from it, i.e., 4.2 and 10.7 BLEU above his reverse-training system, and it remains below it only between Sicilian and Italian. Hence, the low score into Sicilian on our test set is probably due to the test set and to the metric more than to the model. Retrained from scratch with his configuration, the original baseline does not reach its published scores on his data, while our alignment added to his sheets raises it by 4.0 to 6.5 BLEU.

Several extensions are open. In particular, the aligner could merge the short verses of poetry, as simulated in section 4.1, and could collect the Sicilian left unaligned as well as the English, which would enlarge the in-domain pool for back-translation. Also, the training data could be weighted by quality, e.g., favoring the issues published after 1998 or the texts written or translated by Cipolla, whose style varies less than that of the whole journal (the hard part being to identify them, since the issues carry no such metadata). If the back-translations of the web Sicilian turn out to be of little use, forward-translations produced by a small model trained on the literary text are an alternative for the pretraining stage. Furthermore, the forgetting of the untrained directions could be limited by mixing into the adaptation a large multi-parallel corpus (e.g., Bible translations in many directions), at the price of a much longer training. It would also be interesting to compare the encoder representations of translated sentences under bidirectional and unidirectional training, and at different depths, and to measure how much the textbook exercises contribute. Finally, the same pipeline may be applied to varieties close to Sicilian, such as Calabrese and Salentino, whose written texts exhibit similar orthographic diversity. The aligner itself should transfer to other varieties of Italy: on the FLORES-200 devtest, LaBSE retrieves the correct English translation of a Friulian, Ligurian, Lombard, Sardinian or Venetian sentence in 98.5 to 99.9% of the cases (99.0% for Sicilian), and therefore what is missing for them is a bilingual source comparable to *Arba Sicula*.

Our code for the pipeline and the evaluation is publicly available at https://github.com/dmeoli/SicilianNMT, together with a small serving application (a web interface and a Telegram bot, both built on a FastAPI service over the NLLB adapter). The scored back-translations of section 6.3 are released on Hugging Face, one dataset per direction (https://huggingface.co/

datasets/Napizia); the text of the journal is under copyright, and the hand-aligned sheets are not public, so the curated set is not released.

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